



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

PAPER 1 Multiple Choice

6031/1

JUNE 2023 SESSION

1 hour

Additional materials:

Data Booklet

Scientific/Electronic calculator

Multiple Choice answer sheet

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 14 printed pages and 2 blank pages.

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Section A

For each question there are four possible answers, A, B, C and D. Choose the one you consider to be correct.

- 1 A volume of 10 cm^3 of propene was completely burnt in oxygen.

What are the volumes of oxygen gas used and carbon dioxide produced (measured under same conditions of temperature and pressure)?

	oxygen	carbon dioxide
A	30	30
B	45	30
C	45	60
D	90	60

- 2 What is common among the species; NH_4^+ , SO_2 and H_3O^+ ?

- A They all have a planar shape.
- B They are all acidic.
- C They all involve dative bonding.
- D They are all polar.

- 3 Which compound contains particles with a coordination number of 6?

- A diamond
- B sodium chloride
- C silicon dioxide
- D copper

- 4 A mass of 0.50 g of a volatile liquid was introduced into a globe of 1000 cm^3 capacity. The globe was heated to 91°C so that all the liquid vapourised. Under these conditions, the vapour exerted a pressure of 0.25 atmospheres .

Which expression gives the relative molecular mass of the liquid?

A
$$\frac{0.50 \times 8.314 \times (91 + 273)}{101325 \times 0.25 \times 1000 \times 10^{-3}}$$

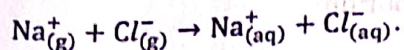
B
$$\frac{0.50 \times 8.314 \times (91 + 273)}{0.25 \times 1000 \times 10^{-3}}$$

C
$$\frac{0.50 \times 8.314 \times 91}{0.25 \times 1000}$$

D
$$\frac{0.50 \times 8.314 \times (91 + 273)}{101325 \times 0.25 \times 1000 \times 10^{-6}}$$



- 5 A chemical change is represented by the equation shown:

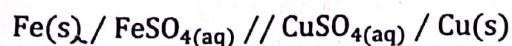


The chemical change shows

- A standard enthalpy of hydration of both $\text{Na}_{(\text{g})}^{+} + \text{Cl}_{(\text{g})}^{-}$
 - B lattice enthalpy of $\text{Na}_{(\text{g})}^{+}\text{Cl}_{(\text{s})}^{-}$
 - C enthalpy change of solution of $\text{Na}_{(\text{g})}^{+}\text{Cl}_{(\text{s})}^{-}$
 - D first ionisation energy of Na and first electron affinity of Cl
- 6 Which one ion has the smallest enthalpy change of hydration.

- A Al^{3+}
- B Mg^{2+}
- C Ca^{2+}
- D Pb^{2+}

- 7 Which one is the correct standard e.m.f. of the cell?



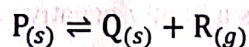
- A -0.10 V
- B -0.78 V
- C $+0.10 \text{ V}$
- D $+0.78 \text{ V}$

- 8 A mass of 0.60 g of silver were deposited in 60 minutes using a current of 0.30 A.

The quantity of charge needed to deposit 1 mole of silver is

- A 12 C.
- B 720 C.
- C 1 296 C.
- D 129 600 C.

- 9 A heterogeneous chemical equilibrium is represented by



The equilibrium constant, K_p , is

- A $\frac{(\text{P}_{\text{P}})}{(\text{P}_{\text{Q}})(\text{P}_{\text{R}})}$
- B $\frac{(\text{P}_{\text{Q}})(\text{P}_{\text{R}})}{(\text{P}_{\text{P}})}$
- C (P_{R})
- D $\frac{(\text{P}_{\text{Q}})^2}{(\text{P}_{\text{P}})}$



- 10 Solubility product of PbSO_4 in water is $1.6 \times 10^{-8} \text{ mol}^2 \text{ dm}^{-6}$.

What is the solubility of PbSO_4 in $0.1 \text{ mol dm}^{-3} \text{ Pb(NO}_3)_2$?

- A $1.00 \times 10^{-1} \text{ mol dm}^{-3}$
- B $1.60 \times 10^{-7} \text{ mol dm}^{-3}$
- C $4.00 \times 10^{-5} \text{ mol dm}^{-3}$
- D $4.00 \times 10^{-4} \text{ mol dm}^{-3}$

- 11 Which equation explains why liquid ammonia conducts electricity?

- A $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$
- B $2\text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{NH}_2^-$
- C $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$
- D $\text{NH}_4^+ + \text{OH}^- \rightleftharpoons \text{NH}_3 + \text{H}_2\text{O}$

- 12 Which ion catalyses the reaction between $\text{I}^-_{(\text{aq})}$ and $\text{S}_2\text{O}_8^{2-}_{(\text{aq})}$?

- A Cr^{3+}
- B Zn^{2+}
- C Cu^{2+}
- D Co^{3+}

- 13 Which chloride is a volatile liquid and readily hydrolyses in water?

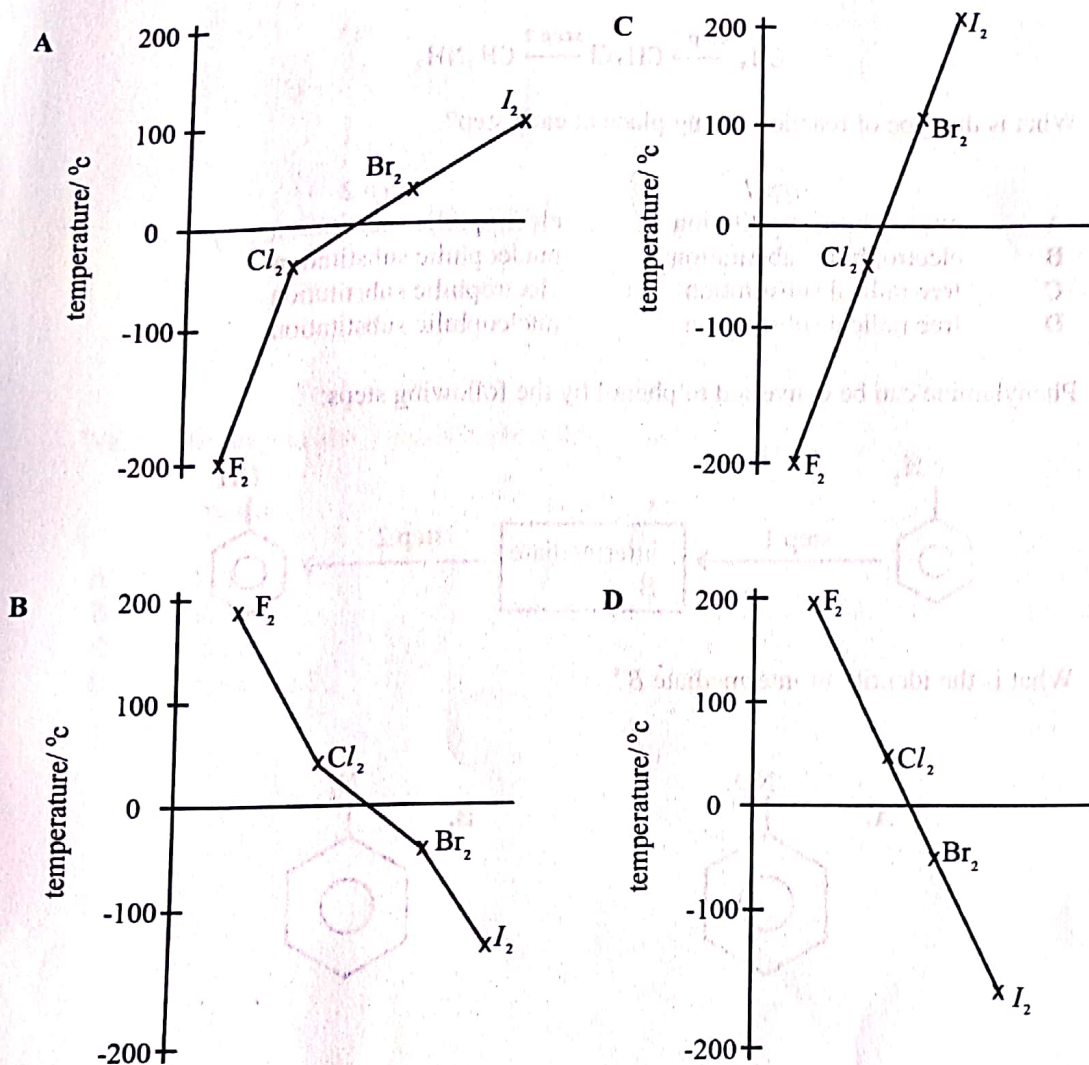
- A AlCl_3
- B CCl_4
- C SiCl_4
- D PCl_5

- 14 Why is beryllium fluoride soluble whereas calcium fluoride is insoluble in water?

- A Ca^{2+} has a greater polarising power.
- B Beryllium fluoride is ionic whereas calcium fluoride is covalent.
- C Lattice dissociation enthalpy of beryllium fluoride is less exothermic than that of calcium fluoride.
- D The small Be^{2+} ion has a highly negative enthalpy of hydration.



- 15 Which graph shows the correct trend in the boiling points of halogens.

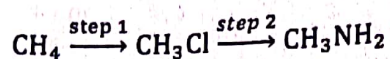


- 16 What is the oxidation state of chromium in $[Cr(H_2O)_4Cl_2]^+ Cl^- \cdot 2H_2O$?

- A +6
B +3
C +2
D +1



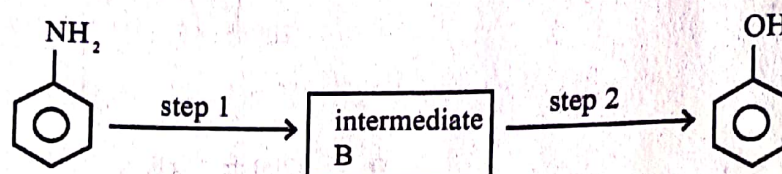
- 17 The reaction scheme shows how amines can be synthesized.



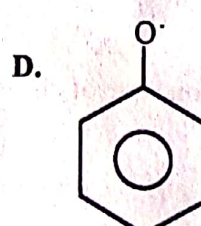
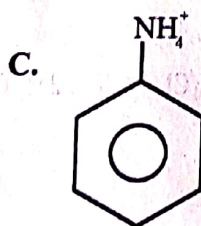
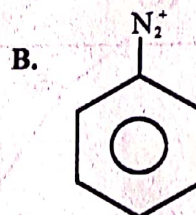
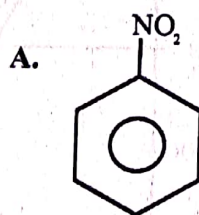
What is the type of reaction taking place at each step?

- | | <i>step 1</i> | <i>step 2</i> |
|---|----------------------------|----------------------------|
| A | nucleophilic substitution | electrophilic substitution |
| B | electrophilic substitution | nucleophilic substitution |
| C | free radical substitution | electrophilic substitution |
| D | free radical substitution | nucleophilic substitution |

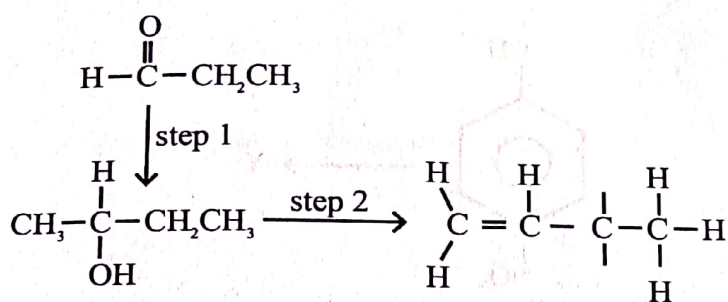
- 18 Phenylamine can be converted to phenol by the following steps:



What is the identity of intermediate B?



- 19 The reaction scheme shows the formation of but-1-ene:

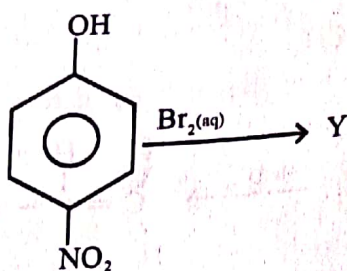


What is the reagent and condition needed for step 2?

	reagent	condition
A	KCN	reflux
B	LiAlH_4	dry ether
C	KCN	ethanol
D	conc H_2SO_4	180°C



20 A reaction scheme for 4-nitrophenol is shown:

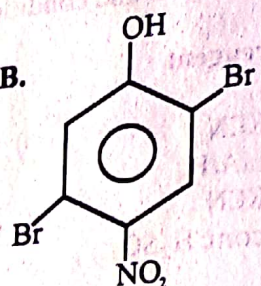


What is the structure of Y?

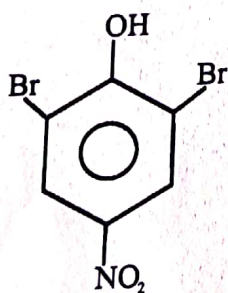
A.



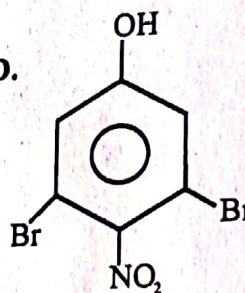
B.



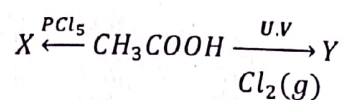
C.



D.



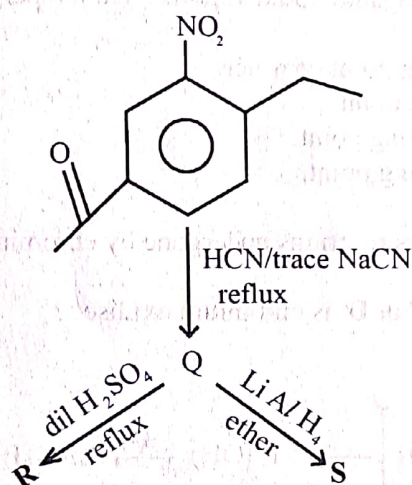
- 21 Ethanoic acid can be converted to X and Y as shown.



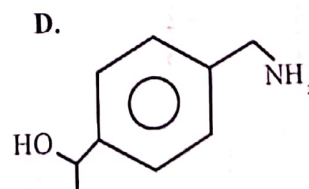
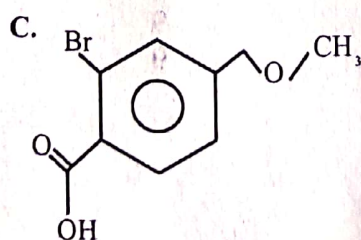
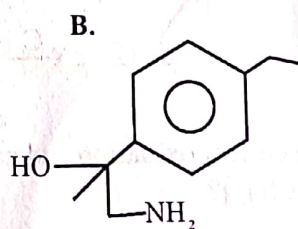
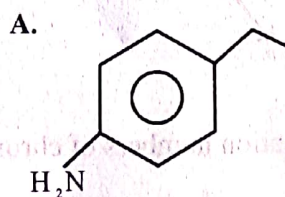
What is the possible molecular formulae of X and Y?

- | | X | Y |
|---|-------------|--------------|
| A | CH_3COCl | $CH_2ClCOOH$ |
| B | CH_3COCl | CH_3COCl |
| C | KCN | CH_3COCl |
| D | CH_3CCl_3 | $CH_2ClCOCl$ |

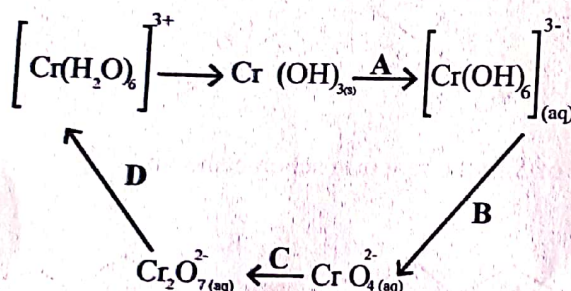
- 22 The reaction scheme shows how compounds Q, R and S can be synthesized.



What is the structure of S in the reaction scheme shown?



- In which reaction, A, B, C or D, is chromium oxidised?



- | | oxidation number | coordination number |
|----------|------------------|---------------------|
| A | +3 | 6 |
| B | +2 | 6 |
| C | +2 | 9 |
| D | +3 | 9 |



28 A heterogeneous catalyst is a catalyst that

- A stays the same throughout the reaction.
- B is a different state to the reactants.
- C changes throughout the reaction.
- D is the same state as the reactants.

29 Which pollutant leads to the formation of acid rain?

- A carbon monoxide
- B chlorofluorocarbons
- C tetraethyllead
- D sulphur dioxide

30 An organic compound **P** of M_r 56, reduces Fehling's solution and reacts with H_2 in the presence of a nickel catalyst. **P** contains 64.3% C; 7.1% H and 28.6% O by mass.

The structural formula of **P** is

- A $CH_3CH_2(OH).$
- B $CH_3CH_2COH.$
- C $CH_2CHCOH.$
- D $CH_3 - C - CH_3.$
 $\parallel$
 O



Section B

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct. (You may find it helpful to put a tick against the statement(s) which you consider to be correct).

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 Which oxide(s) has/have a giant molecular structure?

1. SiO_2
2. PbO
3. CO_2

32 Which one is/are the correct property/properties of anhydrous aluminium chloride?

1. acidic in water
2. vapour has a covalent molecular structure
3. solid is ionic

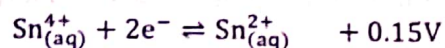
33 The K_b value of a $5.00 \times 10^{-3} \text{ mol dm}^{-3}$ solution of methylamine is $4.17 \times 10^{-4} \text{ mol dm}^{-3}$.

Which statement(s) is/are true about the solution?

1. $[\text{OH}] = 1.44 \times 10^{-3}$
2. pH is 11.15
3. is more basic than $5.00 \times 10^{-3} \text{ mol dm}^{-3}$ ammonia.



- 34 The standard potential for the electrode reaction is represented by the equation.



What may be deduced from the information?

1. tin is more stable in the +4 than in the +2 oxidation state
2. tin (IV) compounds can act as oxidising agents
3. SnO_2 decomposes to SnO

- 35 Anhydrous calcium nitrate and anhydrous magnesium nitrate both decompose on heating, producing nitrogen dioxide and oxygen.

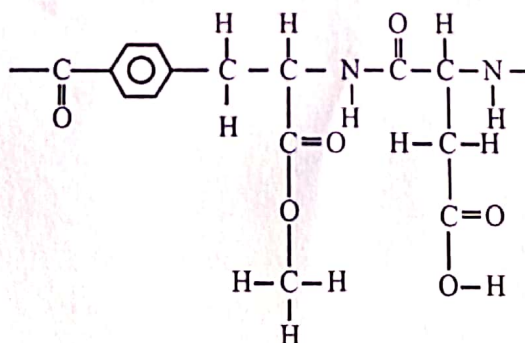
Which statement(s) about the two reactions is/are correct?

1. nitrogen dioxide and oxygen produced from magnesium nitrate are at lower temperature than that of calcium nitrate.
2. the volume of oxygen produced is 25% the volume of nitrogen dioxide produced for both nitrates.
3. both nitrates leave a residue of metal oxide.

- 36 Which reagent(s) can be used to distinguish between propanone and propanol?

1. Tollens reagent
2. Fehling's reagent
3. 2,4 - dinitrophenylhydrazine

- 37 The structure of an organic compound is shown.



The structure shown is/can

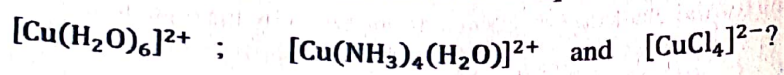
1. a condensation polymer.
2. a copolymer.
3. undergo saponification.

6031/1 J2023

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38 Which statement(s) is/are true about the complexes.



1. They all exhibit conformational isomerism.
2. The ligands form covalent bond with copper ion.
3. Oxidation state of copper is +2.

39 Natural resources can be conserved by

1. using crude oil as a source of fuel.
2. reusing waste material.
3. recycling non-biodegradable waste material.

40 Nanomaterials exist in the form of

1. suspensions.
2. colloids.
3. Solids.

